

AP Flex Score Methodology

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1 Introduction

This paper outlines the method used to calculate an AP Flex Score. It is by no means a complete documentation of the app. Rather, it is a snapshot into the inner workings of the application.

1.1 Terminology

Term Used	Definition
Aggregate Score	The final Flex Score given. On the certificate, it's the largest number
Raw Score	A preliminary Flex Score, without any external buffs or debuffs
Buff	Positive multipliers that increase the Flex Score
Debuff	Negative multipliers that decrease the Flex Score

Table 1: Terminology

2 Raw Score

The Raw Flex score is calculated with scoring curves from previous AP exams, in addition to the perceived difficulty of the programs by students.

2.1 Variables Used

Symbol	Definition
Variables & Functions	
$s(a)$	Flex score of AP exam a
$s_e(a)$	Score buff/debuff from exam score
a_d	The base difficulty of an AP exam $\in [1, 10]$
a_s	Score of an AP exam $\in [1, 5]$
$a[i]$	The portion of people who obtain a score of i or higher. $i \in [1, 5], a[i] \in [0, 1]$

Table 2: Variables

2.2 Scoring for an Individual Exam

The scoring function for an individual test uses a base difficulty score for each test, in addition to a multiplier based on the percentile scored on the test (or 1.0 if scores have not been released yet).

AP Test	Base Difficulty (a_d)
Computer Science A	4.2
Macroeconomics	4.9
English Language	5.2
Calculus BC	5.9
US History	6.0
Biology	6.4
Physics 1	7.2
Physics C EM	8.0

Table 3: Base Difficulties of Select AP Tests

A complete set of difficulties can be found here.

In addition, common tests not yet on the ranking were added by hand. Although these additions were not performed as rigorously as the initial scoring, they provide an approximation before a true survey can be undertaken.

Additional AP Tests	Base Difficulty (a_d)	Justification
Research	5.0	Difficulty equal to Seminar
Precalculus	4.0	Early exam taken by many

Table 4: Base Difficulties of Additional AP Tests

2.3 Omitted Exams

Some tests were also omitted due to the risk of under/over representing difficulty. These tests fell into three main categories.

1. Language based ones (Chinese, Spanish, French) that may have difficulty curves skewed by native speakers taking the exam
2. New/Pilot exams with too few data points to determine difficulty.
3. Tests with too few takers. For example, AP tests such as Latin with only 4000 takers could not have their difficulty accurately gauged.

2.4 Adjusting for Exam Score

Based on varying difficulties to get a certain score, buffs and debuffs are then applied. The formula is as follows:

$$s_e(a) = \left(\sum_{j=1}^{a_s-1} 2 \times a[j] \right) + a[a_s]$$

A Note on Notation: $\times$ will be used to denote multiplication, while $\cdot$ will be used to denote a dot product

Using this formula, the following distributions are generated

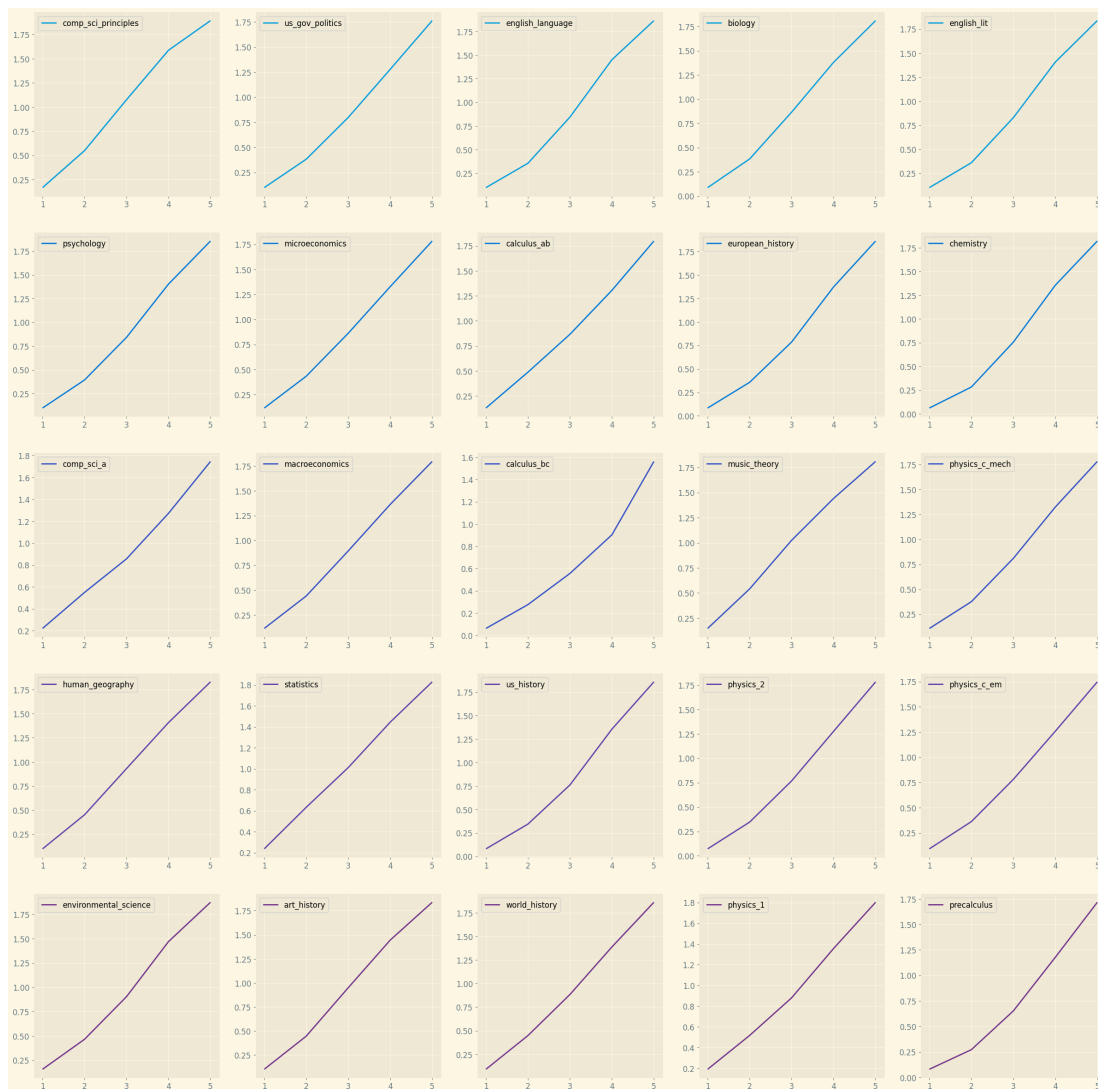


Figure 1: Distributions for 25 APs

2.5 Combining Models

Finally, both models are combined to create an aggregate Flex Score for each AP. This is done by multiplying the base difficulty by the exam score adjustment.

$$s(a) = a_d \times s_e(a)$$

3 Similarity Debuffs

Taking many similar AP Tests is less difficult than taking many widely different tests, often due to overlapping course material or concepts. Thus, to reflect this, a similarity debuff was created.

3.1 Variables Used

Symbol	Definition
Variables & Functions	
$c_{sim}(A, B)$	Cosine Similarity of vectors A and B
$c_{nr sim}(A, B)$	Normalized Cosine Similarity of vectors A and B
$d(a, b)$	The percent debuff of AP tests a , and b , in decimal form
$s(a)$	Flex Score of AP Exam a (From Section 2)
$d_q(a, b)$	Quantity to debuff by, generated from AP a and b
a_E	The vector embedding of an AP exam
h_{sim}	Highest pairwise similarity between two tests
l_{sim}	Lowest pairwise similarity between two tests
Notation	
M_i	Item i of vector M
$M \cdot N$	The dot product of vectors M and N
$ M $	The norm of vector M , computed as $\sqrt{\sum_{i=1}^n (M_i)^2}$

Table 5: Variables

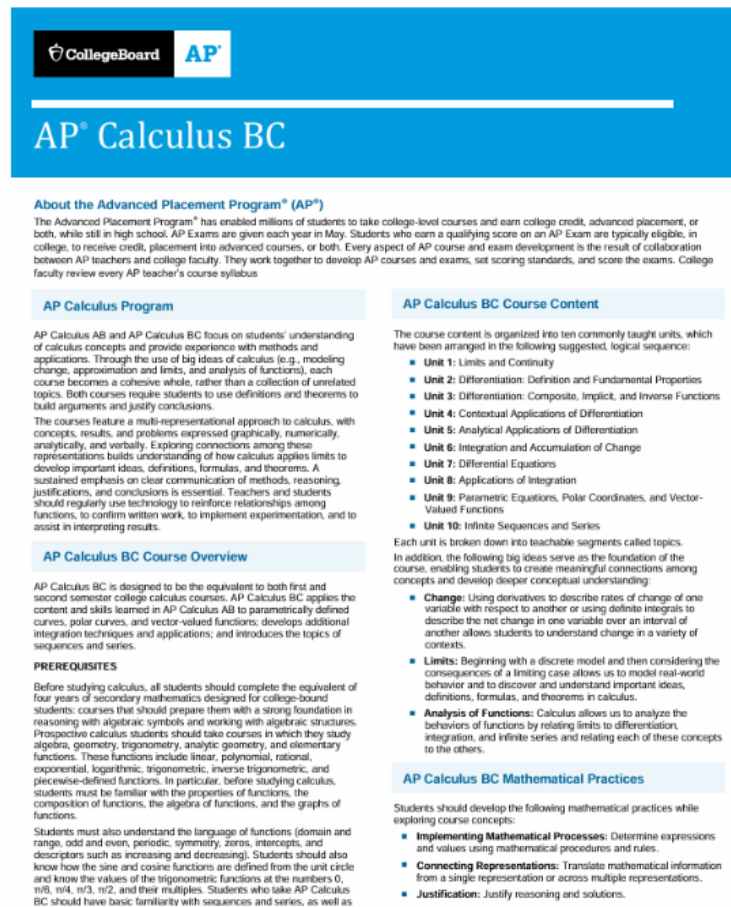
3.2 Creating the Embeddings

To find the similarity of AP tests, an embedding of each AP was first created. This was done with **all-MiniLM-L6-v2** available in the Python Sentence Transformers module.

```
from sentence_transformers import SentenceTransformer

model = SentenceTransformer("sentence-transformers/all-MiniLM-L6-v2")
```

The text itself used in the embedding was a course overview document provided by the College Board. This document was chosen since it encapsulated enough data about each course to be meaningful, without being bloated with boilerplate.



The image shows the AP Calculus BC Course Overview document from CollegeBoard. It includes the AP logo and title, a section about the Advanced Placement Program (AP), and detailed course content for AP Calculus BC. The course content is organized into ten units: Unit 1: Limits and Continuity, Unit 2: Differentiation: Definition and Fundamental Properties, Unit 3: Differentiation: Composite, Implicit, and Inverse Functions, Unit 4: Contextual Applications of Differentiation, Unit 5: Analytical Applications of Differentiation, Unit 6: Integration and Accumulation of Change, Unit 7: Differential Equations, Unit 8: Applications of Integration, Unit 9: Parametric Equations, Polar Coordinates, and Vector-Valued Functions, and Unit 10: Infinite Sequences and Series. It also lists mathematical practices: Implementing Mathematical Processes, Connecting Representations, and Justification.

Figure 2: AP Calculus BC Course Overview

3.3 Using the Embeddings

To calculate the similarity between two AP tests, Cosine Similarity is used. This formula is commonly used in processes such as Retrieval Augmented Generation and compares the "directions" of two vectors rather than their "distance" (as with Euclidian Similarity).

$$c_{sim}(A, B) = \frac{A \cdot B}{\|A\| \times \|B\|}$$

3.4 Calculating Debuff

With a method of finding similarity between two AP tests now complete, all that remains is to calculate the debuff. To do this, the similarity of exams is first min-max normalized to further minimize any effects of boilerplate.

$$c_{nrsim}(A, B) = \frac{c_{sim}(A, B) - l_{sim}}{h_{sim} - l_{sim}}$$

Since the debuff should increase exponentially in respect to similarity, to penalize highly similar courses while reducing penalty in marginal similar courses, exponential functions are used.

$$d(a, b) = \frac{e^{4.2 \times c_{nrsim}(a_E, b_E)} - 1}{e^{4.2} - 1}$$

This function creates the following curve:

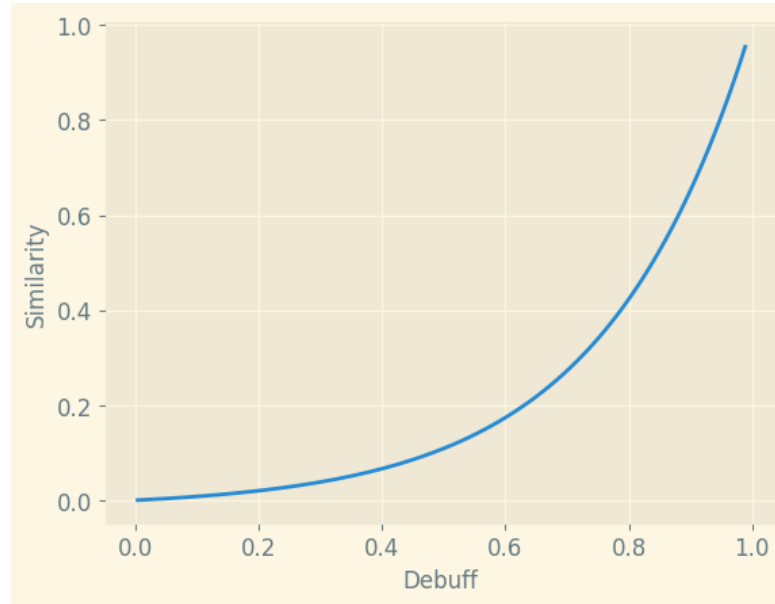


Figure 3: Debuffs Curve

3.5 Applying Debuffs

To apply debuffs, $d(a, b)$ is first calculated, then applied on the lower score of the two. This is since the perceived impressiveness of a given AP test is not diminished by the completion of other tests below it, but only by tests above it.

Furthermore, a limit of $d(a, b) \geq 0.05$ is placed to avoid any trivial debuff.

Mathematically, for two AP tests, a and b

$$d_q(a, b) = \begin{cases} s(b) \times (1 - d(a, b)) & \text{if } s(a) < s(b) \\ s(a) \times (1 - d(a, b)) & \text{if } s(a) \geq s(b) \end{cases}$$

4 Buffs

Not all AP test takers are built equal. For example, taking three Advanced Placement exams in Freshman year is much more impressive than taking the three same exams in Senior year. Furthermore, other factors like self studying or a rare Perfect Score may change the impressiveness of a given exam.

4.1 Age-based Buffs

These buffs are applied to AP exams taken before Junior year (11th grade), where most people take their exams. The following buffs are available.

Year of Exam	Multiplier
Middle School (Any year j 9th grade)	1.15 (+15%)
Freshman Year	1.05 (+5%)
Sophomore Year	1.02 (+2%)

Table 6: Age Based Buffs

Note that these multipliers are automatically applied in the web app.

4.2 Other Buffs

In addition to rewarding age, buffs are also given to reward self studying and rare Perfect Scores, two respectable accomplishments that increase the perceived impressiveness of an AP.

Buff Name	Multiplier
Self Study	1.1 (+10%)
Partial Self Study	1.05 (+5%)
Perfect Score	2.5 (+150%)

Table 7: Other Buffs

Notes:

1. The Self Study Buff is for cases where no formal tutoring was used, whether inside of school or outside of school. (Note that using resources such as prep books and online videos is allowed)
2. The Partial Self Study Buff is to be used when the AP Course was taken outside of school. A good way to differentiate between this and the Self Study Buff is whether or not payment was made for services.

3. The Perfect Score buff is not for a 5. Rather, it is for the case where every point possible has been earned on the exam. This usually is disclosed in a letter sent by College Board.

5 Putting Everything Together

With the separate sections of the model complete, all that remains is to integrate them.

5.1 Variables Used

Symbol	Definition
Variables & Functions	
$s(a)$	Raw flex score of AP exam a (Section 2)
$d_q(a, b)$	Quantity to debuff by, generated from AP a and b (Section 3)
A	Set of AP Exams taken
s_r	Raw flex score of all AP exams taken
s_b	Flex score of all AP exams taken with buffs
s_d	Similarity Debuff sum
s_f	Final Flex Score
Notation	
A_i	The i^{th} AP taken
$ A $	Cardinality (Length) of set A

Table 8: Variables

5.2 Calculating the Raw Flex Score

We first set the Raw Flex Score, s_r , to be equal to the sum of all individual raw flex scores.

$$s_r = \sum_{i=1}^{|A|} s(A_i)$$

5.3 Calculating the Buffs

Next, buffs are applied. This is to keep consistent with the principle that similarity debuffs only diminish the impressiveness of the least impressive exam. Since things such as self studying factor into the impressiveness of an exam, they are applied first.

We set s_b to the buffed Flex Score, and modify it in the following ways:

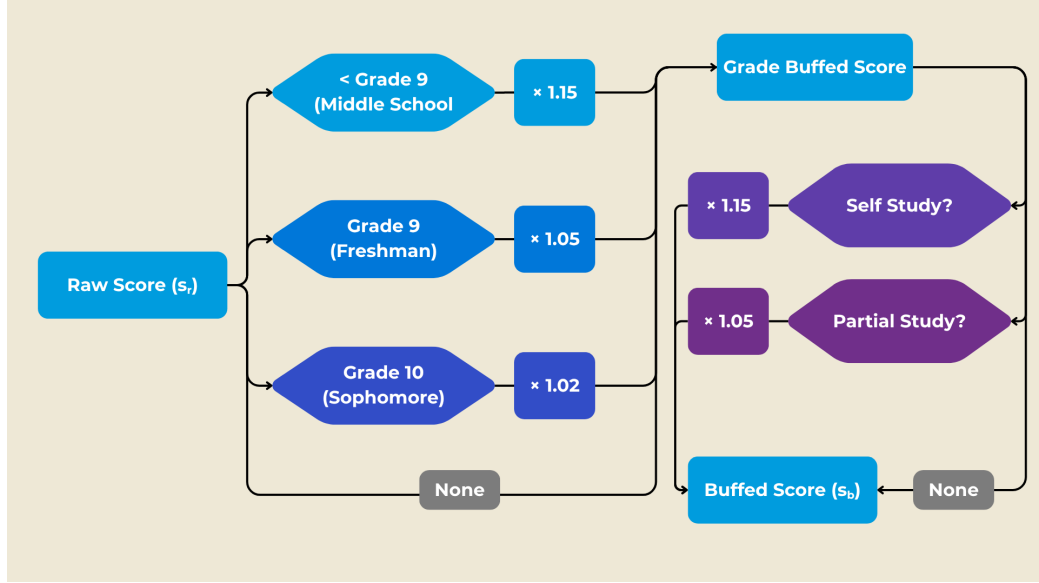


Figure 4: Flowchart for Buff Application

5.4 Similarity Debuffs

To calculate similarity debuffs, we sum up the pairwise debuffs for each set of APs.

$$s_r = \sum_{i=1}^{|A|} \sum_{j=i+1}^{|A|} d_q(A_i, A_j)$$

By using two pointers to iterate through, pairwise debuffs are not counted more than once per pair.

5.5 Final Score

To calculate the final Flex Score, the total debuff is subtracted from the Buffered Flex Score.

$$s_f = s_b - s_d$$

This concludes the calculation of the Flex Score.

6 Appendix

Included in the appendix are code snippets and bits of data used to build the model, so that someone can recreate the pipeline to a good degree of accuracy if ever needed.

A note on AI Generation: AI was used to create some of the longer lists and visualizations seen. Any AI generated/assisted content is marked with a red asterisk (*)

6.1 Links to Documents Used for Embedding

* AI Generated list with human edits

```

json_name_to_link = {
    "comp_sci_principles": "https://apcentral.collegeboard.org/media/pdf/ap-computer-science-principles-course-overview.pdf",
    "psychology": "https://apcentral.collegeboard.org/media/pdf/ap-psychology-course-overview.pdf",
    "comp_sci_a": "https://apcentral.collegeboard.org/media/pdf/ap-computer-science-a-course-overview.pdf",
    "human_geography": "https://apcentral.collegeboard.org/media/pdf/ap-human-geography-course-overview.pdf",
    "environmental_science": "https://apcentral.collegeboard.org/media/pdf/ap-environmental-science-course-overview.pdf",
    "microeconomics": "https://apcentral.collegeboard.org/media/pdf/ap-microeconomics-course-overview.pdf",
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    "statistics": "https://apcentral.collegeboard.org/media/pdf/ap-statistics-course-overview.pdf",
    "art_history": "https://apcentral.collegeboard.org/media/pdf/ap-art-history-course-overview.pdf",
    "english_language": "https://apcentral.collegeboard.org/media/pdf/ap-english-language-and-composition-course-overview.pdf",
    "calculus_ab": "https://apcentral.collegeboard.org/media/pdf/ap-calculus-ab-course-overview.pdf",
    "calculus_bc": "https://apcentral.collegeboard.org/media/pdf/ap-calculus-bc-course-overview.pdf",
    "us_history": "https://apcentral.collegeboard.org/media/pdf/ap-us-history-course-at-a-glance.pdf",
    "world_history": "https://apcentral.collegeboard.org/media/pdf/ap-world-history-modern-course-overview.pdf",
    "biology": "https://apcentral.collegeboard.org/media/pdf/ap-biology-course-overview.pdf",
    "european_history": "https://apcentral.collegeboard.org/media/pdf/ap-european-history-course-overview.pdf",
    "music_theory": "https://apcentral.collegeboard.org/media/pdf/ap-music-theory-course-overview.pdf",
    "physics_1": "https://apcentral.collegeboard.org/media/pdf/ap-physics-1-course-overview.pdf",
    "physics_2": "https://apcentral.collegeboard.org/media/pdf/ap-physics-2-course-overview.pdf",
    "english_lit": "https://apcentral.collegeboard.org/media/pdf/ap-english-literature-and-composition-course-overview.pdf",
    "chemistry": "https://apcentral.collegeboard.org/media/pdf/ap-chemistry-course-overview.pdf",
    "precalculus": "https://apcentral.collegeboard.org/media/pdf/ap-precalculus-course-overview.pdf",
    "seminar": "https://apcentral.collegeboard.org/media/pdf/ap-seminar-course-overview.pdf",
    "research": "https://apcentral.collegeboard.org/media/pdf/ap-research-course-overview.pdf",
    "spanish_language": "https://apcentral.collegeboard.org/media/pdf/ap-spanish-language-and-culture-course-overview.pdf",
    "us_gov_politics": "https://apcentral.collegeboard.org/media/pdf/ap-united-states-government-and-politics-course-overview.pdf",
    "us_comparative_gov": "https://apcentral.collegeboard.org/media/pdf/ap-comparative-government-and-politics-course-overview.pdf",
    "physics_c_mech": "https://apcentral.collegeboard.org/media/pdf/ap-physics-c-mechanics-course-overview.pdf",
    "physics_c_em": "https://apcentral.collegeboard.org/media/pdf/ap-physics-c-electricity-and-magnetism-course-overview.pdf"
}

```

6.2 20 Lowest and Highest Pairwise Similarities (Normalized)

```
[['APs us_history and physics_c_em', 0.0],
 ['APs comp_sci_a and us_history', 0.019544823],
 ['APs comp_sci_principles and us_history', 0.020003228],
 ['APs calculus_ab and us_history', 0.041081373],
 ['APs us_history and music_theory', 0.04196005],
 ['APs us_history and chemistry', 0.045450337],
 ['APs calculus_bc and us_history', 0.05443318],
 ['APs us_history and physics_c_mech', 0.05722298],
 ['APs us_history and precalculus', 0.06452961],
 ['APs us_history and physics_2', 0.06706035],
 ['APs psychology and us_history', 0.07021109],
 ['APs us_history and us_gov_politics', 0.07172862],
 ['APs environmental_science and us_history', 0.081032224],
 ['APs us_history and research', 0.0860037],
 ['APs us_history and biology', 0.105443366],
 ['APs us_history and physics_1', 0.10792627],
 ['APs us_history and seminar', 0.1080124],
 ['APs statistics and us_history', 0.14139114],
 ['APs microeconomics and us_history', 0.1604751],
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 ['APs psychology and research', 0.7230907],
 ['APs calculus_ab and precalculus', 0.72804374],
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 ['APs physics_2 and physics_c_em', 0.8283438],
 ['APs english_language and english_lit', 0.86785775],
 ['APs comp_sci_principles and comp_sci_a', 0.89053786],
 ['APs us_gov_politics and us_comparative_gov', 0.89822936],
 ['APs physics_1 and physics_2', 0.90067357],
 ['APs physics_1 and physics_c_mech', 0.91785556],
 ['APs microeconomics and macroeconomics', 0.9349662],
 ['APs physics_2 and physics_c_mech', 0.9353477],
 ['APs seminar and research', 0.95586294],
 ['APs world_history and european_history', 0.99878705],
 ['APs calculus_ab and calculus_bc', 1.0]]
```

6.3 Visualization of AP Clustering

* AI Generated visualization code

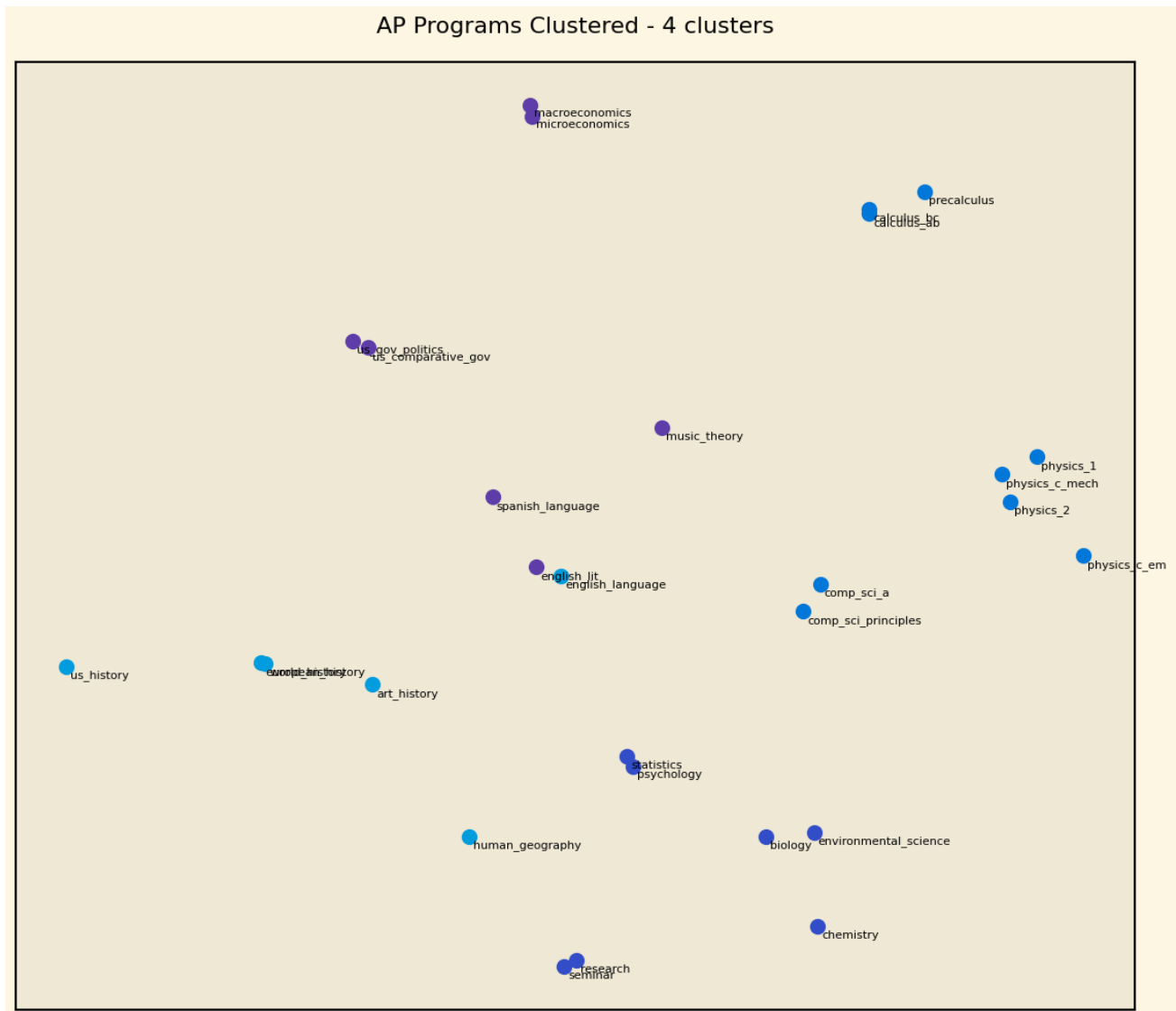


Figure 5: APs Clustered - 4